Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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Abstract

[To be continued. . .]

- Appendix on s-wave SC ghost simulations on HM
- Appendix on triplet pairing in MFT self-consistent scheme
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List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ-wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
iHF	Iterative Hartree-Fock
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Introduction

This thesis project is about my favorite ice cream flavor. [To be continued. . .]

Chapter 1

Mean-field theory discussion of the EHM

The general idea of this project is to apply HF methods to the EHM, derive analytically the approximated quadratic hamiltonian of Eq. (**??**) and then apply the iHF procedure sketched in Sec. **??** to extract the self-consistent values of the various HFPs. One of the advantages of using iHF methods is that we can impose rather simply the symmetries of the HF approximated wavefunction.

This is most useful when dealing with competing phases. A phase transition is basically described by some spontaneous symmetry breaking (SSB) and the rising to a non-zero value of an order parameter. We will deal with three phases:

1. the normal phase, where no symmetry is broken;
2. the antiferromagnetic phase, where we allow for spin density to be non-uniform across sites;
3. the superconducting phase, where we do not conserve particle number.

Each phase is discussed in a separate chapter. Many other "phases" are possible, based on which symmetries are respected and which are not.

For each phase, we will extract the best approximation to the true ground state wavefunction. In computational terms, we will find three solutions to the general self-consistency problem of Eq. (**??**). In the end we want to compare the free energy of these three solutions, assessing which one is the most stable. Any HF approximation, being by construction not the true ground state, overestimates energy. Thus, comparing energies of our solutions we get a hint of the features of the true ground state.

1.1 The EHM symmetries

In this section we present the symmetries of the extended Hubbard model

$$\hat{H}_{\mathrm{EHM}} = -t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma} + U\sum_{i}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow} - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{n}_{i\sigma}\hat{n}_{j\sigma'}$$

The symmetries of the model can be divided in two groups: the ones related to the geometry of the square lattice, and those related to the algebraic properties of $\hat{H}_{\mathrm{EHM}}$. We start by the lattice symmetries.

1.1.1 Lattice symmetries

We study the EHM on the planar square lattice, with one orbital per site. A sketch of the lattice $\mathcal{L}$ is given in Fig. [Missing]. We indicate with L_x the number of sites in direction x, and L_y in direction y. The hamiltonian of Eq. (**??**) respects three fundamental lattice symmetries:

1. Discrete translations symmetry in both the x and y directions. We refer to the groups of one-site translations as as $\mathrm{T}^{(x)}_1$, $\mathrm{T}^{(x)}_1$. It holds:

$$\mathrm{T}^{(\ell)}_n \supset \mathrm{T}^{(\ell)}_{kn} \qquad k\in\mathbb{N} \tag{1.1}$$

A shift in direction ℓ by kn sites can be decomposed in k shift by n sites. Invariance by n-sites shifts, thus, implies invariance also for kn-sites shifts.

2. Discrete $\pi/2$ rotations symmetry. This symmetry is described by the four-fold cyclic point group C_4, containing rotations by angle $\pi/2, \pi, 3\pi/2$ and identity. A similar property as for the translations holds for cyclic point groups,

$$\mathrm{C}_n \supset \mathrm{C}_{n/k} \qquad k, n/k \in \mathbb{N} \tag{1.2}$$

Invariance for rotations of angle $2\pi/n$ implies invariance for rotations of angle $k \times 2\pi/n$.

3. Inversion symmetry. We define I_2 as the inversion group in two-dimensional space, containing the reflection operator and identity. This point may appear redundant, because

$$\mathrm{I}_2 \equiv \mathrm{C}_2 \tag{1.3}$$

In 2D space, to rotate by π around the origin and to mirror the position of a point are the same operation.

Although not a symmetry, it is worth mentioning that the planar square lattice is bipartite. Let us give a definition. Consider a graph $\mathcal{L}$ and the set of its vertices $V(\mathcal{L}) = \{v_i\}$. Let $E(\mathcal{L})$ be the set of graph edges,

$$E(\mathcal{L}) = \{(v, w) \mid v, w \in V(\mathcal{L}) \wedge v,\, w \text{ are connected}\}$$

The graph is said to be bipartite if two sub-graphs $\mathcal{S}_1$, $\mathcal{S}_2$ exist such that:

- The vertices of $\mathcal{L}$ split in the two sub-graphs,

$$V(\mathcal{L}) = V(\mathcal{S}_1) \cup V(\mathcal{S}_2)$$

- Vertices of one sub-graph are linked only to vertices of the other sub-graph. In other words

$$v, w \in \mathcal{S}_i \quad \Longrightarrow \quad (v, w) \notin E(\mathcal{L})$$

The planar square lattice is bipartite: it can be divided in two square lattices, as represented in Fig. [Missing]. Two sites are "connected" if linked by a non-zero hopping operator. Which is, if they are NNs.

1.1.2 Spin and charge symmetries

The conventional HM shows a global U(2) symmetry, extended to $\mathrm{SU}(2) \times \mathrm{SU}(2)/\mathbb{Z}_2 = \mathrm{SO}(4)$ symmetry at half-filling on a bipartite lattice [9], such as the square lattice. For the EHM, we limit to prove that we can identify a $\mathrm{U}(1) \times \mathrm{SU}(2)$.

Gauge symmetry. The EHM is symmetric under the global transformation

$$\hat{c}_{i\sigma} \to e^{-i\eta}\hat{c}_{i\sigma} \qquad \hat{c}^\dagger_{i\sigma} \to e^{i\eta}\hat{c}^\dagger_{i\sigma} \qquad \eta \in \mathbb{R} \tag{1.4}$$

To prove this, it suffices to note that

$$\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'} \quad \to \quad e^{i\eta}\hat{c}^\dagger_{i\sigma}e^{-i\eta}\hat{c}_{j\sigma'} = \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'}$$

As a consequence, any product of fermionic operators with an equal number of annihilations and creations is symmetric. Hence the kinetic part $\hat{H}_t$ and the interactions $\hat{H}_U$, $\hat{H}_V$ are invariant. As a consequence, the total number of particles (charge)

$$\hat{N} = \sum_{i\sigma} \hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}$$

is a good quantum number. The symmetry is described by the transformation of Eq. (1.4), which is a pure phase shift by η. The relative group is $\mathrm{U}^{\mathrm{c}}(1)$, with a superscript "c" indicating "charge".

Global spin symmetry The EHM is described by three operators: NN hopping, independent of σ; local repulsion between opposite-spin densities; non-local attraction, independent of σ. If we rotate all spins by an equal amount, the hamiltonian must remain identical. An identical observation holds for the number operator, $\hat{N}$. Thus $\hat{H}_{\mathrm{EHM}} - \mu\hat{N}$ must be exhibit a global SU(2) symmetry, related to the rotation in 3D space of all spins altogether.

Let us prove this rigorously. Let the on-site spinor:

$$\hat{\Psi}_i \equiv \begin{bmatrix} \hat{c}_{i\uparrow} \\ \hat{c}_{i\downarrow} \end{bmatrix}$$

and the Pauli matrices

$$\tau^x \equiv \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \qquad \tau^y \equiv \begin{bmatrix} & -i \\ i & \end{bmatrix} \qquad \tau^z \equiv \begin{bmatrix} 1 & \\ & -1 \end{bmatrix}$$

We define the local spin operator as:

$$\hat{S}_i^\alpha \equiv \frac{1}{2}\hat{\Psi}_i^\dagger \tau^\alpha \hat{\Psi}_i \qquad \left[\hat{S}_j^\alpha, \hat{S}_j^\beta\right] = i\sum_\gamma \epsilon_{\alpha\beta\gamma}\hat{S}_j^\gamma$$

with $\epsilon_{\alpha\beta\gamma}$ the Levi-Civita tensor. It can be checked by explicit calculation that the $\mathfrak{su}(2)$ spin algebra is actually respected. For example, consider $\alpha = x$, $\beta = y$:

$$\begin{aligned}
\left[\hat{S}_j^x, \hat{S}_j^y\right] &= \frac{1}{4}\left[\hat{\Psi}_j^\dagger \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \hat{\Psi}_j, \hat{\Psi}_j^\dagger \begin{bmatrix} & -i \\ i & \end{bmatrix} \hat{\Psi}_j\right] \\
&= \frac{1}{4}\left[\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow} + \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow}, -i\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow} + i\hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow}\right] \\
&= \frac{i}{4}\left[\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}, \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow}\right] - \frac{i}{4}\left[\hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow}, \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}\right] \\
&= \frac{i}{2}\left[\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}, \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow}\right] \\
&= \frac{i}{2}\left(\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}\hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow} - \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow}\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}\right) \\
&= \frac{i}{2}\left(-\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\downarrow}\hat{c}_{j\uparrow} + \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\uparrow} + \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\uparrow}\hat{c}_{j\downarrow} - \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\downarrow}\right) \\
&= \frac{i}{2}\left(\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\uparrow} - \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\downarrow}\right) \\
&= i\hat{S}_j^z
\end{aligned}$$

We used Fermi anticommutation rules, $\{\hat{c}_{i\sigma}, \hat{c}_{j\sigma'}^\dagger\} = \delta_{ij}\delta_{\sigma\sigma'}$. Identical derivations can be carried out for all other components.

Let the raising and lowering operators:

$$\hat{S}_j^+ \equiv \frac{\hat{S}_j^x + i\hat{S}_j^y}{2} \qquad \hat{S}_j^- \equiv \frac{\hat{S}_j^x - i\hat{S}_j^y}{2}$$

and the global spin operators

$$\hat{S}^\alpha \equiv \sum_{i\in\mathcal{L}} \hat{S}_i^\alpha \qquad \left[\hat{S}^\alpha, \hat{S}^\beta\right] = i\sum_\gamma \epsilon_{\alpha\beta\gamma}\hat{S}^\gamma$$

being $\mathcal{L}$ the lattice. Explicitly,

$$\hat{S}^z \equiv \frac{1}{2}\sum_{i\in\mathcal{L}}(\hat{n}_{i\uparrow} - \hat{n}_{i\downarrow}) \qquad \hat{S}^+ \equiv \frac{1}{2}\sum_{i\in\mathcal{L}} \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\downarrow} \qquad \hat{S}^- \equiv \left(\hat{S}^+\right)^\dagger$$

The EHM hamiltonian and the number operator commutate with the global spin operator. For the kinetic part $\hat{H}_t$ and the local repulsion $\hat{H}_U$ this is a well known result [**pavarini2015many**, 9],

$$\left[\hat{H}_t + \hat{H}_U, \hat{\mathbf{S}}\right] = 0 \qquad \left[\hat{N}, \hat{\mathbf{S}}\right] = 0$$

For the non-local attraction, since it depends solely on total densities on neighbouring sites, spin-rotational invariance must hold. Anyway, let us give a short formal proof following this chain of results:

1. The non-local attraction can be written as:

$$\begin{aligned} \hat{H}_V &= V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} \\ &= V \sum_{\langle ij \rangle} \hat{\Psi}_i^\dagger \hat{\Psi}_i \hat{\Psi}_j^\dagger \hat{\Psi}_j \end{aligned} \tag{1.5}$$

   having we used $\hat{\Psi}_i^\dagger \hat{\Psi}_i = \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} + \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow}$.

2. Similarly, the number operator is given by:

$$\begin{aligned} \hat{N} &= \sum_{i \in \mathcal{L}} \sum_{\sigma} \hat{n}_{i\sigma} \\ &= \sum_{i \in \mathcal{L}} \hat{\Psi}_i^\dagger \hat{\Psi}_i \end{aligned} \tag{1.6}$$

3. Let now $\hat{R}(\boldsymbol{\theta})$ be any global spin rotation,

$$\hat{R}(\boldsymbol{\theta}) \equiv \exp\left(-i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}\right) \quad \text{where} \quad \boldsymbol{\theta} = (\theta_x, \theta_y, \theta_z)$$

   Spinors transform as:

$$\hat{R}(\boldsymbol{\theta}) \hat{\Psi}_i \hat{R}(\boldsymbol{\theta})^\dagger = \mathcal{R}(\boldsymbol{\theta}) \hat{\Psi}_i$$

   with $\mathcal{R}(\boldsymbol{\theta}) \in \mathrm{SU}(2)$ the unitary matrix associated to said rotation,

$$\mathcal{R}(\boldsymbol{\theta}) = \exp\left(-\frac{i}{2} \boldsymbol{\theta} \cdot \boldsymbol{\tau}\right) \quad \text{where} \quad \boldsymbol{\tau} = (\tau_x, \tau_y, \tau_z) \tag{1.7}$$

   Let us prove this. Let $\hat{A} = i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}$ e $\hat{B} = \hat{\Psi}_i$. The following identity, a consequence of Baker-Campbell-Housdorff formula, holds:

$$e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + \left[\hat{A}, \hat{B}\right] + \frac{1}{2!}\left[\hat{A}, \left[\hat{A}, \hat{B}\right]\right] + \frac{1}{3!}\left[\hat{A}, \left[\hat{A}, \left[\hat{A}, \hat{B}\right]\right]\right] + \cdots$$

   The first order of the expansion gives:

$$\begin{aligned} [i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, \hat{\Psi}_i] &= i \sum_{\alpha} \theta_\alpha [\hat{S}^\alpha, \hat{\Psi}_i] \\ &= -\frac{i}{2} \sum_{\alpha} \theta_\alpha \tau^\alpha \hat{\Psi}_i \\ &= -\frac{i}{2} \boldsymbol{\theta} \cdot \boldsymbol{\tau} \hat{\Psi}_i \end{aligned}$$

   We omitted some purely algebraic passages. The second order:

$$\begin{aligned} \left[i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, \left[i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, \hat{\Psi}_i\right]\right] &= \left[i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, -\frac{i}{2}(\boldsymbol{\theta} \cdot \boldsymbol{\tau})\hat{\Psi}_i\right] \\ &= -\frac{i}{2} \boldsymbol{\theta} \cdot \boldsymbol{\tau} \left[i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, \Psi_i\right] \\ &= \frac{1}{4} (\boldsymbol{\theta} \cdot \boldsymbol{\tau})^2 \hat{\Psi}_i \end{aligned}$$

Phase	**SSBs**
AF	$\mathrm{T}_1^{(\ell)} \to \mathrm{T}_2^{(\ell)}$
	$\mathrm{SU}(2) \to \mathrm{U}^z(1)$
SC	$\mathrm{U}^{\mathrm{c}}(1) \to 0$

Table 1.1 SSB interpretation of the AF and SC phases. In the last line, the 0 on the SSB right-hand side indicates a completely disrupted symmetry. MOVE ME

and so on. Collecting all terms,

$$
\begin{aligned}
\hat{R}(\boldsymbol{\theta})\hat{\Psi}_i\hat{R}(\boldsymbol{\theta})^\dagger &= \hat{\Psi}_i - \frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\hat{\Psi}_i + \frac{1}{2!\times 4}(\boldsymbol{\theta}\cdot\boldsymbol{\tau})^2\hat{\Psi}_i + \cdots \\
&= \sum_{k=1}^{\infty}\frac{1}{k!}\left(-\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\right)^k\hat{\Psi}_i \\
&= \mathcal{R}(\boldsymbol{\theta})\hat{\Psi}_i
\end{aligned}
$$

In last passage, we identified the series as the Taylor expansion of the exponential, recognising the expression for $\mathcal{R}(\boldsymbol{\theta})$ given in Eq. (1.7).

4. Then, the group acts on the product $\hat{\Psi}_i^\dagger\hat{\Psi}_i$ as

$$
\begin{aligned}
\hat{R}\hat{\Psi}_i^\dagger\hat{\Psi}_i\hat{R}^\dagger &= \hat{R}\hat{\Psi}_i^\dagger\hat{R}^\dagger\hat{R}\hat{\Psi}_i\hat{R}^\dagger \\
&= \hat{\Psi}_i^\dagger\mathcal{R}^\dagger\mathcal{R}\hat{\Psi}_i \\
&= \hat{\Psi}_i^\dagger\hat{\Psi}_i
\end{aligned}
$$

being $\mathcal{R} \in \mathrm{SU}(2)$. Recalling Eqns. (1.5), (1.6), both $\hat{H}_V$ and $\hat{N}$ are SU(2) symmetric.

The EHM is symmetric under global spin rotations. Note that $\hat{H}_V$ is actually symmetric under local spin rotations, meaning that we could rotate the spin at each site by a different amount and get the same interaction. The same holds for $\hat{H}_U$. It is the kinetic operator that makes the EHM globally (and not locally) SU(2) symmetric.

Note that the $U^{\mathrm{c}}(1)$ gauge symmetry and the SU(2) spin rotations symmetry can be combined,

$$
\mathrm{U}^{\mathrm{c}}(1) \otimes \mathrm{SU}(2) = \mathrm{U}(2)
$$

We will refer to U(2) symmetry to indicate full charge-spin conservation. Finally, note the group of 3D rotations contains the subgroup of 2D rotations around a given axis z,

$$
\mathrm{U}^z(1) \subset \mathrm{SU}(2)
$$

where the z superscript serves to distinguish from the gauge group. The z axis does not need to be identified with the "real" z axis, perpendicular with the lattice plane. Any given direction works. Inverting our logic, we can build the SU(2) generators, starting from $\hat{S}^z$, after we have chosen the direction z. In terms of commutation relations, a given operator can respect $\mathrm{U}^z(1)$ but not SU(2): in that case, it will commutate to zero with $\hat{S}^z$, and not the other spin operators.

1.1.3 Particle-hole symmetry at half-filling

The half-filled EHM on the square lattice exhibits particle-hole symmetry (PHS). The presence of this additional symmetry is a consequence of the fact that the square lattice is bipartite. Let us prove this. Consider the generic unitary operation

$$
\mathrm{P} \quad : \quad \hat{c}_{i\sigma} \quad \to \quad e^{i\eta_{i\sigma}}\hat{h}_{i\sigma}^\dagger \tag{1.8}
$$

being in general η_i a phase dependent on the fermionic quantum state $(i\sigma)$. Inserting the above transformation in the EHM, we get

$$
\mathrm{P}\{\hat{H}\} = -t\sum_{\langle ij\rangle}\sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma})}\hat{h}_{i\sigma}\hat{h}_{j\sigma}^\dagger + U\sum_i \hat{h}_{i\uparrow}\hat{h}_{i\downarrow}\hat{h}_{i\downarrow}^\dagger\hat{h}_{i\uparrow}^\dagger - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}_{j\sigma'}^\dagger\hat{h}_{i\sigma}^\dagger
$$

For the two interactions $\hat{H}_U$, $\hat{H}_V$ the phase factors cancelled out. We aim to find the conditions under which the above hamiltonian is mapped onto the EHM of (**??**),

$$\mathrm{P}\{\hat{H}\} \stackrel{?}{=} -t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{h}^{\dagger}_{i\sigma}\hat{h}_{j\sigma} + U\sum_{i\in\mathcal{L}}\hat{h}^{\dagger}_{i\uparrow}\hat{h}^{\dagger}_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}^{\dagger}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma}$$

Before starting the derivation, which is an extension of the argument of Arovas et al. [9] about the conventional HM, let us discuss some properties of the hole operators. Due to the presence of the phase factors in the PHS operative definition, the Fermi anticommutation rules hold regardlessly of the phase factor:

$$\begin{aligned}\{\hat{h}_{i\sigma}, \hat{h}^{\dagger}_{j\sigma'}\} &= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})}\{\hat{c}^{\dagger}_{i\sigma}, \hat{c}_{j\sigma'}\}\\ &= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})}\delta_{ij}\delta_{\sigma\sigma'}\\ &= \delta_{ij}\delta_{\sigma\sigma'}\end{aligned}$$

The on-site density operator is given by:

$$\hat{h}_{i\sigma}\hat{h}^{\dagger}_{i\sigma} = \hat{n}_{i\sigma}$$

being $\hat{n}_{i\sigma}$ the fermionic number operator on site i with spin σ. It follows, for quartic interactions

$$\begin{aligned}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^{\dagger}_{j\sigma'}\hat{h}^{\dagger}_{i\sigma} &= -\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^{\dagger}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'}\\ &= \hat{h}_{i\sigma}\hat{h}^{\dagger}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^{\dagger}_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'}\\ &= -\hat{h}^{\dagger}_{i\sigma}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^{\dagger}_{j\sigma'} + \hat{h}_{j\sigma'}\hat{h}^{\dagger}_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'}\\ &= \hat{h}^{\dagger}_{i\sigma}\hat{h}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'}\hat{h}_{j\sigma'} - \hat{h}^{\dagger}_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^{\dagger}_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'}\\ &= -\hat{h}^{\dagger}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'} - \hat{h}^{\dagger}_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^{\dagger}_{j\sigma'} - \left(\hat{h}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'} - \hat{h}^{\dagger}_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'}\\ &= \hat{h}^{\dagger}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} + (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) - 1 - \left(\hat{h}_{i\sigma}\hat{h}^{\dagger}_{j\sigma'} - \hat{h}^{\dagger}_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'}\end{aligned} \tag{1.9}$$

The first term of the last passage is mirrored with respect to the starting term. Now, let us deal with the kinetic, local and non-local interactions separately.

Kinetic term. Consider the kinetic term,

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma})}\hat{h}_{i\sigma}\hat{h}^{\dagger}_{j\sigma}$$

Anti-commutate the two fermionic operators, and exchange site labels $i \leftrightarrow j$. We get:

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma}+\pi)}\hat{h}^{\dagger}_{i\sigma}\hat{h}_{j\sigma}$$

A necessary condition on the unitary operation P is the following,

$$\eta_{i\sigma} + \eta_{j\sigma} = (2k-1)\pi \quad \text{with} \quad k \in \mathbb{Z} \tag{1.10}$$

valid for NN sites.

Local repulsion. Substitute Eq. (1.9) in $\hat{H}_U$,

$$\begin{aligned}\mathrm{P}\{\hat{H}_U\} &= U\sum_{i\in\mathcal{L}}\hat{h}_{i\uparrow}\hat{h}_{i\downarrow}\hat{h}^{\dagger}_{i\downarrow}\hat{h}^{\dagger}_{i\uparrow}\\ &= U\sum_{i\in\mathcal{L}}\hat{h}^{\dagger}_{i\uparrow}\hat{h}^{\dagger}_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U\sum_{i\in\mathcal{L}}(\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}) - U\sum_{i\in\mathcal{L}}(1)\\ &= U\sum_{i}\hat{h}^{\dagger}_{i\uparrow}\hat{h}^{\dagger}_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U(\hat{N} - L_xL_y)\end{aligned}$$

For the half-filled state $\langle\hat{N}\rangle = L_xL_y$. The unitary map of Eq. (1.8), independently of the condition of Eq. (1.10), leaves the hamiltonian identical at half-filling.

Symmetry	Operations
$\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$	One-site discrete translations separately in both directions
C_4	Four-fold point group (real $\pi/2$ angle rotations)
$\mathrm{U}^{\mathrm{c}}(1)$	Charge-conserving global phase shifts
$\mathrm{SU}(2)$	Three-dimensional rotations in global spin space

Table 1.2 Model symmetries and relative group operations. Note that $\mathrm{U}^{\mathrm{c}}(1)$ represents the gauge symmetry linked to charge conservation.

Non-local attraction. Proceed identically for $\hat{H}_V$,

$$\begin{aligned}
\mathrm{P}\{\hat{H}_V\} &= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}^\dagger_{j\sigma'} \hat{h}^\dagger_{i\sigma} \\
&= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}^\dagger_{i\sigma} \hat{h}^\dagger_{j\sigma'} \hat{h}_{j\sigma'} \hat{h}_{i\sigma} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) + V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} (1) \\
&= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}^\dagger_{i\sigma} \hat{h}^\dagger_{j\sigma'} \hat{h}_{j\sigma'} \hat{h}_{i\sigma} - 2zV(\hat{N} - L_x L_y)
\end{aligned}$$

with $z = 4$ the lattice coordination number. In the last passage, we used that on the square lattice

$$\begin{aligned}
\sum_{\langle ij \rangle} (1) &= \sum_{i \in \mathcal{L}/2} z \\
&= \frac{L_x L_y}{2} \times z
\end{aligned} \tag{1.11}$$

As before, the non-local attraction is particle-hole symmetric at half-filling. As for the local repulsion, PHS is given just the operation of Eq. (1.8). The condition of Eq. (1.10) is irrelevant.

Recollecting everything, if P is defined by the unitary operation

$$\mathrm{P} \quad : \quad \begin{cases} \hat{c}_{i\sigma} \to \hat{h}^\dagger_{i\sigma} & \text{for} \quad i \in \mathcal{S}_1 \\ \hat{c}_{i\sigma} \to -\hat{h}^\dagger_{i\sigma} & \text{for} \quad i \in \mathcal{S}_2 \end{cases}$$

and the system is half-filled, PHS is respected. Any other phase, as long as it alternates by π on the two sublattices, is good enough. As for the conventional HM [9], this result can be generalised to any bipartite lattice. In turn, if a lattice is not bipartite, PHS is not present. To summarise, we report in Tab. 1.2 all identified symmetries.

1.2 Symmetry breaking channels in real space

In many-body Physics, a phase transition is described by some order parameter rising from zero to a non-zero value. This is associated with some SSB. In order to simulate different phases, we need a way to distinguish them based on which symmetries they break (or not). We also need a way to work computationally in a given symmetry sector. HF methods, by reducing the hamiltonian to a non-interacting approximation, offer a simple framework to these ends. Indeed, once we identified all symmetries of a given phase, we just impose them on our approximation of the full hamiltonian. The found ground-state approximation will entail these symmetries by construction.

Recall now the discussion of the HF method of Sec. **??**. The ending point were Eqns. (**??**), (**??**), which identify the self-consistent single-particle fields which approximate the original quartic hamiltonian. In order to get there, we applied Wick's theorem to the quartic part of the hamiltonian and derived the most general expressions of $\mathbf{h}$. In next sections here apply Wick's theorem to the quartic operators of $\hat{H}_{\mathrm{EHM}}$ and discuss the symmetry properties of the resulting decomposition.

Before starting, let us define precisely the three phases we will investigate in terms of symmetries:

- Normal phase: no broken symmetry.

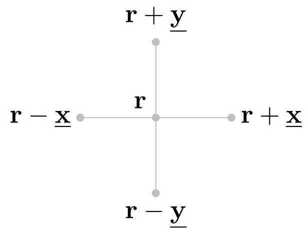


Figure 1.1 | Schematic representation of the four NNs of a given site i for a planar square lattice.

- Antiferromagnetic phase: simultaneous $\mathrm{SU}(2) \to \mathrm{U}^z(1)$ and $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)} \to \mathrm{T}_2^{(x)} \otimes \mathrm{T}_2^{(y)}$ SSBs. Among all possible realisations of antiferromagnetism, we realise a SDW with alternating distribution of spin densities;
- Superconducting phase: simultaneous $\mathrm{U}^c(1) \to 0$ and $\mathrm{C}_4 \to \mathrm{C}_2$ SSBs. We investigate anisotropic superconductivity, allowing for the particle number to be not conserved and for nematicity.

We will theoretically discuss the nematic SSB $\mathrm{C}_4 \to \mathrm{C}_2$ also for normal and AF phases, as if discrete rotational symmetry was broken. As we will see, while it not trivial to exclude it *a priori*, it will actually turn out that the HF ground state preserves C_4 in both cases.

We now move to a more detailed discussion of each hamiltonian contribution. In next sections, we switch from the site index i to the vector notation $\mathbf{r}$, indicating the site 2D position on the lattice

$$i \quad \to \quad \mathbf{r} = (x, y)$$

with $x, y \in \mathbb{N}$. The lattice versors are:

$$\underline{\mathbf{x}} \equiv (1, 0) \qquad \underline{\mathbf{y}} \equiv (0, 1)$$

as in Fig. 1.1.

1.2.1 Local repulsion

The local repulsion $\hat{H}_U$ is the quartic operator of the conventional HM,

$$\begin{aligned} \hat{H}_U &= U \sum_{\mathbf{r} \in \mathcal{L}} \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}\downarrow} \\ &= U \sum_{\mathbf{r} \in \mathcal{L}} \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}^\dagger_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \end{aligned}$$

In order to obtain the self-consistent values of $h^{+-}_{\alpha\beta}$, $h^{++}_{\alpha\beta}$ we need to compute an expectation value of the full hamiltonian. Wick's theorem guarantees that, for the wavefunctions we are using, the expectation value of a quartic fermionic operator is the sum of all fully contracted terms. Recalling the nomenclature of Eq. (**??**),

$$\langle \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}^\dagger_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle = \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}_{\mathbf{r}\uparrow} \rangle \langle \hat{c}^\dagger_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\downarrow} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}_{\mathbf{r}\downarrow} \rangle \langle \hat{c}^\dagger_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}^\dagger_{\mathbf{r}\downarrow} \rangle \langle \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle}_{\text{Cooper}} \tag{1.12}$$

where the equal sign holds as long as we perform the averaging operation on the subspace of **gaussian states**. Now we discuss the above EVs separately, identifying if any can be neglected based on selection rules in a given phase.

Hartree term. The Hartree part of the above decomposition is a product of density EVs,

$$\langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle$$

Recalling the groups of Tab. 1.2, we see that the density operator preserves all symmetries. Thus no selection rule applies, and the above EVs are in principle non-zero.

Term	Normal phase	Antiferromagnet	Superconductor
Hartree			
Fock	Prohibited	Prohibited	Prohibited
Cooper	Prohibited	Prohibited	

Table 1.3 Summary of Wick's terms selection rules and relevant contributions to the effective hamiltonian for each phase limitedly to $\hat{H}_U$.

Fock terms. For the Fock part, notice that

$$\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow} = \hat{S}^+_{\mathbf{r}}$$

being $S^+_{\mathbf{r}}$ the local raising operator. From SU(2) algebra we have:

$$\left[\hat{S}^+_{\mathbf{r}}, \hat{S}^\alpha\right] \neq 0 \qquad \alpha \in \{x, y, z\}$$

If the ground-state is invariant under global spin rotations, these EVs must be zero.

Cooper terms. The Cooper part can be non-zero only if charge is not conserved. In terms of the gauge symmetry, it suffices to apply the map of Eq. (1.4) to $\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}$ to conclude that this operator is not gauge invariant.

In Tab. 1.3 we collect the selection rules for the Hartree, Fock and Cooper terms of Eq. (1.12), based on the symmetry specifications given for each phase in above paragraphs.

1.2.2 Non-local attraction

Let us start from the real space form of Eq. (1.31). To carry out a summation over nearest neighbors $\langle\mathbf{r}\mathbf{r}'\rangle$ of a square lattice means precisely to sum over all links of the lattice. Then we can identify the generic opposite-spin (o.s.) term $\hat{H}_V^{\sigma\overline{\sigma}}$ as the one collecting the σ operators of sublattice $\mathcal{S}_a$ and $\overline{\sigma}$ operators of sublattice $\mathcal{S}_b$. The o.s. non-local interactions can be written as a sum of terms over just one of the two sublattices $\mathcal{S}_a$ and $\mathcal{S}_b$. Define the site hamiltonian

$$\hat{h}_V^{(\mathbf{r})} \equiv -V \sum_{\boldsymbol{\delta}} \left(\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow}\right) \quad \text{with} \quad \boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\} \tag{1.13}$$

Then, it follows,

$$\begin{aligned} \hat{H}_V^{(\text{o.s.})} &= \overbrace{\sum_{\mathbf{r}\in\mathcal{S}_a} \hat{h}_V^{(\mathbf{r})}}^{\hat{H}_V^{\uparrow\downarrow}} + \overbrace{\sum_{\mathbf{r}\in\mathcal{S}_b} \hat{h}_V^{(\mathbf{r})}}^{\hat{H}_V^{\downarrow\uparrow}} \\ &= \sum_{\mathbf{r}\in\mathcal{S}} \hat{h}_V^{(\mathbf{r})} \end{aligned} \tag{1.14}$$

Here the notation of Fig. ?? is used. For each site $\mathbf{r}$ in a given sublattice, the nearest neighbors sites are four – all in the other sublattice. Similarly, the same-spin (s.s.) hamiltonian decomposes as

$$\hat{H}_V^{(\text{s.s.})} = -V \sum_{\mathbf{r}\in\mathcal{S}_a} \sum_{\boldsymbol{\delta},\sigma} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma}\right) \quad \text{with} \quad \boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}$$

Opposite-spin terms. Consider first the o.s. terms of Eq. (1.13): take e.g. the NNs term $\hat{n}_{i\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}$. As in Eq. (??), taking Wick's contractions,

$$\begin{aligned} \langle\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle &= \langle\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \\ &\simeq \underbrace{\langle\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle\langle\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}} - \underbrace{\langle\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle\langle\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle}_{\text{Hartree}} \end{aligned}$$

Identical decompositions are given for all others NNs. Of the three terms above:

Sector	Term	Normal phase	Antiferromagnet	Superconductor
o.s.	Hartree			
	Fock	Prohibited	Prohibited	Prohibited
	Cooper	Prohibited	Prohibited	
s.s.	Hartree			
	Fock			
	Cooper	Prohibited	Prohibited	

Table 1.4 Summary of Wick's terms selection rules and relevant contributions to the effective hamiltonian for each phase limitedly to $\hat{H}_V$, assuming preservation of $\mathrm{U}^z(1)$ symmetry.

- The Cooper term breaks $\mathrm{U}^c(1)$ charge symmetry, allowing for supeconducting instabilities;
- The Fock term breaks $\mathrm{U}^z(1)$ symmetry, because it accounts for a site hop *plus* spin flip process;
- The Hartree term can reduce both translational invariance and spin rotational symmetry.

As a general rule of thumb, throughout this text we will discard the o.s. Fock term eveywhere, preserving a robust $\mathrm{U}^z(1)$ symmetry. For the normal state as well as for the AF instability only the Hartree term is to be accounted; for the SC phase also the Cooper term contributes. Note that for superconducting instabilities, due to superexchange mechanism (as explained in App. **??**) the o.s. term accounts for singlet pairing as well as zero-projection triplet pairing. Which channel is preferred, is a matter of thermodynamic advantage.

Same-spin terms. Consider then the same-spin terms: take e.g. the term $\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}$. As above,

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \\ &= \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle\langle \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle}_{\text{Hartree}}\end{aligned}$$

Identical consideration as in the above paragraph hold for each term. The only difference with the o.s. terms is given by the Fock term: since the spin-flip process is absent, now the Fock fluctuations actually contribute as an effective NN hopping term. In other words, this term does not break $\mathrm{U}^z(1)$ symmetry and thus is perfectly legitimate in each phase. Tab. 1.4 summarizes the selection rules hereby presented.

1.3 The EHM in reciprocal space

The EHM is symmetric under discrete lattice translations. We will investigate phases that do not (completely) break translational invariance. It is thus convenient to move to reciprocal space, by Fourier-transforming the hamiltonian. We will use the following convention:

$$\hat{c}_{\mathbf{r}\sigma} \overset{\mathcal{F}}{=} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in\mathrm{BZ}} e^{-i\mathbf{k}\cdot\mathbf{r}} \hat{c}_{\mathbf{k}\sigma} \tag{1.15}$$

Throughout the text, $\overset{\mathcal{F}}{=}$ will indicate a passage where a Fourier transform is used. "BZ" is shorthand notation for the Brillouin Zone.

It is useful to write explicitly two simple properties:

- On the square lattice (as for any bipartite lattice) NNs belong to different sublattices. This implies that any NNs sum can be rewritten as:

$$\sum_{\langle \mathbf{r}\mathbf{r}'\rangle} f(\mathbf{r},\mathbf{r}') = \sum_{\mathbf{r}\in\mathcal{L}/2}\ \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} f(\mathbf{r},\mathbf{r}+\boldsymbol{\delta}) \tag{1.16}$$

with f some function. We are using the notation of Fig. 1.1. The symbol "$\mathcal{L}/2$" indicates one of the two sublattices $\mathcal{S}_1$, $\mathcal{S}_2$ represented in Fig. [Missing];

- The following equation for Kronecker's delta holds:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} e^{i\mathbf{k}\cdot\mathbf{r}} = \delta_{\mathbf{k}=\mathbf{0}} \tag{1.17}$$

On a sublattice:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}/2} e^{i\mathbf{k}\cdot\mathbf{r}} = \frac{1}{2}\delta_{\mathbf{k}=\mathbf{0}} \tag{1.18}$$

Kinetic part. Let us transform $\hat{H}_t$:

$$\begin{aligned}
\hat{H}_t &= -t \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}'\sigma} + \text{h.c.}\right] \\
&\overset{\mathcal{F}}{=} -t \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in\text{BZ}} e^{i\mathbf{k}\cdot\mathbf{r}}\hat{c}^\dagger_{\mathbf{k}\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}'\in\text{BZ}} e^{-i\mathbf{k}'\cdot\mathbf{r}'}\hat{c}_{\mathbf{k}'\sigma} + \text{h.c.} \\
&= -2t \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma} \times \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}/2} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \times \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} e^{i\mathbf{k}'\cdot\boldsymbol{\delta}} \\
&= -2t \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \left[\frac{e^{ik_x}+e^{-ik_x}}{2} + \frac{e^{ik_y}+e^{-ik_y}}{2}\right] \qquad (1.19) \\
&= \sum_{\mathbf{k}\in\text{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \qquad (1.20)
\end{aligned}$$

where we used Eqns. (1.16), (1.18). We defined the bare bands:

$$\epsilon_{\mathbf{k}} \equiv -2t\left(\cos k_x + \cos k_y\right) \tag{1.21}$$

Local repulsion. The local repulsion transforms as:

$$\begin{aligned}
\hat{H}_U &= U \sum_{\mathbf{r}\in\mathcal{L}} \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow} \\
&\overset{\mathcal{F}}{=} U \sum_{\mathbf{r}\in\mathcal{L}} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i\mathbf{k}_1\cdot\mathbf{r}}\hat{c}^\dagger_{\mathbf{k}_1\uparrow} e^{i\mathbf{k}_2\cdot\mathbf{r}}\hat{c}^\dagger_{\mathbf{k}_2\downarrow} e^{-i\mathbf{k}_3\cdot\mathbf{r}}\hat{c}_{\mathbf{k}_3\downarrow} e^{-i\mathbf{k}_4\cdot\mathbf{r}}\hat{c}_{\mathbf{k}_4\uparrow} \\
&= \frac{U}{L_x L_y} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} \hat{c}^\dagger_{\mathbf{k}_1\uparrow}\hat{c}^\dagger_{\mathbf{k}_2\downarrow}\hat{c}_{\mathbf{k}_3\downarrow}\hat{c}_{\mathbf{k}_4\uparrow} \times \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \\
&= \frac{U}{L_x L_y} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} \hat{c}^\dagger_{\mathbf{k}_1\uparrow}\hat{c}^\dagger_{\mathbf{k}_2\downarrow}\hat{c}_{\mathbf{k}_3\downarrow}\hat{c}_{\mathbf{k}_4\uparrow}\delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}
\end{aligned}$$

where we used Eq. (1.17). Let $\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4=\mathbf{K}$, and define $\mathbf{k}$, $\mathbf{k}'$ such that

$$\mathbf{k}_1 \equiv \mathbf{K}+\mathbf{k} \qquad \mathbf{k}_2 \equiv \mathbf{K}-\mathbf{k} \qquad \mathbf{k}_3 \equiv \mathbf{K}-\mathbf{k}' \qquad \mathbf{k}_4 \equiv \mathbf{K}+\mathbf{k}'$$

Sums over these variables must be intended as over the Brillouin Zone (BZ). Hence:

$$\hat{H}_U \overset{\mathcal{F}}{=} \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{1.22}$$

Non-local attraction. To transform the non-local attraction $\hat{H}_V$, it is useful to start by defining:

$$h_{\mathbf{r}\sigma\sigma'} \equiv -V \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma'}\right)$$

with:

$$\hat{H}_V = \sum_{\mathbf{r}\in\mathcal{L}/2}\sum_{\sigma,\sigma'} h_{\mathbf{r}\sigma\sigma'}$$

The product of densities on neighbouring sites transforms as:

$$\begin{aligned}\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'} &= \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\sigma} \\ &\overset{\mathcal{F}}{=} \frac{1}{(L_xL_y)^2}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} e^{\pm i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}\hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}\end{aligned}$$

Then:

$$\begin{aligned}h_{\mathbf{r}\sigma\sigma'} &\overset{\mathcal{F}}{=} -\frac{V}{(L_xL_y)^2}\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \\ &\qquad\qquad \times\left(e^{i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}} + e^{-i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}\right)\hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma} \\ &= -\frac{2V}{(L_xL_y)^2}\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}}\cos\left[(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}\right]\hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}\end{aligned}$$

The full non-local interaction is given by summing over all sites of $\mathcal{L}/2$ (which is, one sublattice). Applying Eq. (1.18) gives back momentum conservation

$$\frac{1}{L_xL_y}\sum_{\mathbf{r}\in\mathcal{L}/2} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} = \frac{1}{2}\delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}$$

where we used Eq. (1.18). Define $\mathbf{k}$, $\mathbf{k}'$ and $\mathbf{K}$ as we did fo the local repulsion, and

$$\delta\mathbf{k} \equiv \mathbf{k}-\mathbf{k}'$$

Hence:

$$\begin{aligned}\hat{H}_V &= \sum_{\mathbf{r}\in\mathcal{L}/2}\sum_{\sigma,\sigma'} h_{\mathbf{r}\sigma\sigma'} \\ &\overset{\mathcal{F}}{=} -\frac{V}{L_xL_y}\sum_{\sigma\sigma'}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\cos\left(\delta\mathbf{k}\cdot\boldsymbol{\delta}\right)\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \\ &= -\frac{V}{L_xL_y}\sum_{\sigma\sigma'}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\end{aligned} \tag{1.23}$$

$$\begin{aligned}\hat{H}_V \overset{\mathcal{F}}{=} &\overbrace{-\frac{V}{L_xL_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}^{\text{s.s.}} \\ &\underbrace{-\frac{V}{L_xL_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}_{\text{o.s.}}\end{aligned}$$

The o.s. part can be simplified: since

$$\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} = \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}$$

due to fermionic anti-commutation rules, and since the composite transform

$$\mathbf{k}\to-\mathbf{k} \qquad \mathbf{k}'\to-\mathbf{k}'$$

leaves the form factor $\cos(\delta k_x) + \cos(\delta k_y)$ invariant, we reduce the spin sum to two times the $\sigma = \uparrow$ part,

$$\hat{H}_V^{(\text{o.s.})} = -\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)]\, \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{1.24}$$

whiel the explicit s.s. sector is given immediately by:

$$\begin{aligned} \hat{H}_V^{(\text{s.s.})} = -\frac{V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \\ \times \left[\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\uparrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\uparrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} + \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\downarrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\downarrow} \right] \end{aligned} \tag{1.25}$$

The result here obtained will be used various times in next chapters. Note that different Wick contraction schemes will be referred to as:

$$c^\dagger_{\mathbf{K}+\mathbf{k}\sigma} c^\dagger_{\mathbf{K}-\mathbf{k}\sigma'} c_{\mathbf{K}-\mathbf{k}'\sigma'} c_{\mathbf{K}+\mathbf{k}'\sigma} \qquad \text{Cooper contraction} \tag{1.26}$$

$$c^\dagger_{\mathbf{K}+\mathbf{k}\sigma} c^\dagger_{\mathbf{K}-\mathbf{k}\sigma'} c_{\mathbf{K}-\mathbf{k}'\sigma'} c_{\mathbf{K}+\mathbf{k}'\sigma} \qquad \text{Fock contraction} \tag{1.27}$$

$$c^\dagger_{\mathbf{K}+\mathbf{k}\sigma} c^\dagger_{\mathbf{K}-\mathbf{k}\sigma'} c_{\mathbf{K}-\mathbf{k}'\sigma'} c_{\mathbf{K}+\mathbf{k}'\sigma} \qquad \text{Hartree contraction} \tag{1.28}$$

When contracting in the Fock channel, a minus sign must be included.

Local repulsion. A somewhat identical argument can be carried out for the local interaction,

$$\hat{H}_U = U \sum_{\mathbf{r}\in\mathcal{S}} \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}\downarrow}$$

The only difference here is made by the fact that the interaction is repulsive on-site, thus the final structure factor in the Fourier transform disappears as well as the minus sign,

$$\hat{H}_U \stackrel{\mathcal{F}}{=} \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{1.29}$$

When contracting these fermionic operators, identical considerations as for the non-local counterpart (1.34) hold. Note the similarity of this Fourier transform with the o.s. sector of the non-local attraction, given in Eq. (1.34). With this, we conclude the general introduction to the model and pass to the setup of the Hartree-Fock (HF) algorithm.

Let us now extend the discussion to the EHM and consider the NN non-local attraction,

$$\hat{H}_V \equiv -V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma\sigma'} \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}'\sigma'} \tag{1.30}$$

where we changed notation, and replaced the site index i with its 2D position on the lattice

$$i \quad \to \quad \mathbf{r} = (x, y)$$

Let also the lattice versors,

$$\underline{\mathbf{x}} \equiv (1, 0) \qquad \underline{\mathbf{y}} \equiv (0, 1)$$

as in Fig. 1.1.

Evidently the hamiltonian can be decomposed in various spin terms,

$$\begin{aligned} \hat{H}_V &= \sum_{\sigma\sigma'} \hat{H}_V^{\sigma\sigma'} \\ &= \underbrace{\hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow}}_{\text{Same-spin}} + \underbrace{\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow}}_{\text{Opposite-spin}} \end{aligned} \tag{1.31}$$

The role of said terms is crucial in establishing the pairing channel of the dominant physical phase. As an example, the s.s. terms can be seen as a possible source of triplet pairing superconductivity. Next section are devoted to analyze said terms and derive a simple analytical result in reciprocal space.

1.3.1 Fourier transform of the model

Kinetic part. The kinetic part is the easiest to transform. Apply the definition,

$$
\begin{aligned}
-t\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}'\sigma}+\text{h.c.}\right] &\overset{\mathcal{F}}{=} -t\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\frac{1}{\sqrt{L_xL_y}}\sum_{\mathbf{k}\in\text{BZ}}e^{i\mathbf{k}\cdot\mathbf{r}}\hat{c}^{\dagger}_{\mathbf{k}\sigma}\frac{1}{\sqrt{L_xL_y}}\sum_{\mathbf{k}'\in\text{BZ}}e^{-i\mathbf{k}'\cdot\mathbf{r}'}\hat{c}_{\mathbf{k}'\sigma}+\text{h.c.} \\
&= -2t\sum_{\mathbf{k},\mathbf{k}'}\sum_{\sigma}\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma}\frac{1}{L_xL_y}\sum_{\mathbf{r}\in\mathcal{S}/2}e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}}\left[e^{ik'_x}+e^{-ik'_x}+e^{ik'_y}+e^{-ik'_y}\right] \\
&= \sum_{\mathbf{k}\in\text{BZ}}\sum_{\sigma}\epsilon_{\mathbf{k}}\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}
\end{aligned}
\tag{1.32}
$$

where the bare bands were defined,

$$
\epsilon_{\mathbf{k}} \equiv -2t\left(\cos k_x+\cos k_y\right) \tag{1.33}
$$

Non-local attraction. Consider a generic bond, say, the one connecting NNs $\mathbf{r}$ and $\mathbf{r}\pm\boldsymbol{\delta}$. Then:

$$
\begin{aligned}
\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'} &= \hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}^{\dagger}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\sigma} \\
&\overset{\mathcal{F}}{=} \frac{1}{(L_xL_y)^2}\sum_{\mathbf{k}_1,\cdots,\mathbf{k}_4}e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}}e^{\pm i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}\hat{c}^{\dagger}_{\mathbf{k}_1\sigma}\hat{c}^{\dagger}_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}
\end{aligned}
$$

Then, the interaction at site $\mathbf{r}$, spin σ with its NNs at spin σ' – indicated as $(\mathbf{r}\sigma\sigma')$ – is given by

$$
\begin{aligned}
(\mathbf{r}\sigma\sigma') &= -V\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}}\left[\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}+\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma'}\right] \\
&\overset{\mathcal{F}}{=} -\frac{V}{(L_xL_y)^2}\sum_{\boldsymbol{\delta}}\sum_{\mathbf{k}_1,\cdots,\mathbf{k}_4}e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \\
&\qquad\qquad\times\left(e^{i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}+e^{-i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}\right)\hat{c}^{\dagger}_{\mathbf{k}_1\sigma}\hat{c}^{\dagger}_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma} \\
&= -\frac{2V}{(L_xL_y)^2}\sum_{\boldsymbol{\delta}}\sum_{\mathbf{k}_1,\cdots,\mathbf{k}_4}e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}}\cos\left[(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}\right]\hat{c}^{\dagger}_{\mathbf{k}_1\sigma}\hat{c}^{\dagger}_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}
\end{aligned}
$$

The full non-local interaction is given by summing over all sites of $\mathcal{S}/2$ (which is, one sublattice of $\mathcal{S}$). This gives back momentum conservation,

$$
\frac{1}{L_xL_y}\sum_{\mathbf{r}\in\mathcal{S}/2}e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} = \frac{1}{2}\delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}
$$

Let $\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4=\mathbf{K}$, and define $\mathbf{k}$, $\mathbf{k}'$ such that

$$
\mathbf{k}_1\equiv\mathbf{K}+\mathbf{k}\qquad \mathbf{k}_2\equiv\mathbf{K}-\mathbf{k}\qquad \mathbf{k}_3\equiv\mathbf{K}-\mathbf{k}'\qquad \mathbf{k}_4\equiv\mathbf{K}+\mathbf{k}'\qquad \delta\mathbf{k}\equiv\mathbf{k}-\mathbf{k}'
$$

Sums over these variables must be intended as over the Brillouin Zone (BZ). Hence:

$$
\begin{aligned}
\hat{H}_V &= \sum_{\mathbf{r}\in\mathcal{S}_a}\sum_{\sigma\sigma'}(\mathbf{r}\sigma\sigma') \\
&\overset{\mathcal{F}}{=} -\frac{V}{L_xL_y}\sum_{\sigma\sigma'}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\cos\left(\delta\mathbf{k}\cdot\boldsymbol{\delta}\right)\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \\
&= -\frac{V}{L_xL_y}\sum_{\sigma\sigma'}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}
\end{aligned}
\tag{1.34}
$$

which decomposes in s.s. and o.s. channels as:

$$\hat{H}_V \overset{\mathcal{F}}{=} \overbrace{-\frac{V}{L_xL_y}\sum_\sigma\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}^{\text{s.s.}}$$
$$\underbrace{-\frac{V}{L_xL_y}\sum_\sigma\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}_{\text{o.s.}}$$

The o.s. part can be simplified: since

$$\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} = \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}$$

due to fermionic anti-commutation rules, and since the composite transform

$$\mathbf{k}\to-\mathbf{k} \qquad \mathbf{k}'\to-\mathbf{k}'$$

leaves the form factor $\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)$ invariant, we reduce the spin sum to two times the $\sigma=\uparrow$ part,

$$\hat{H}_V^{(\text{o.s.})} = -\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{1.35}$$

whiel the explicit s.s. sector is given immediately by:

$$\hat{H}_V^{(\text{s.s.})} = -\frac{V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]$$
$$\times\left[\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\uparrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\uparrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow}+\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\downarrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\downarrow}\right] \tag{1.36}$$

The result here obtained will be used various times in next chapters. Note that different Wick contraction schemes will be referred to as:

$$c^\dagger_{\mathbf{K}+\mathbf{k}\sigma}c^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}c_{\mathbf{K}-\mathbf{k}'\sigma'}c_{\mathbf{K}+\mathbf{k}'\sigma} \qquad \text{Cooper contraction} \tag{1.37}$$

$$c^\dagger_{\mathbf{K}+\mathbf{k}\sigma}c^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}c_{\mathbf{K}-\mathbf{k}'\sigma'}c_{\mathbf{K}+\mathbf{k}'\sigma} \qquad \text{Fock contraction} \tag{1.38}$$

$$c^\dagger_{\mathbf{K}+\mathbf{k}\sigma}c^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}c_{\mathbf{K}-\mathbf{k}'\sigma'}c_{\mathbf{K}+\mathbf{k}'\sigma} \qquad \text{Hartree contraction} \tag{1.39}$$

When contracting in the Fock channel, a minus sign must be included.

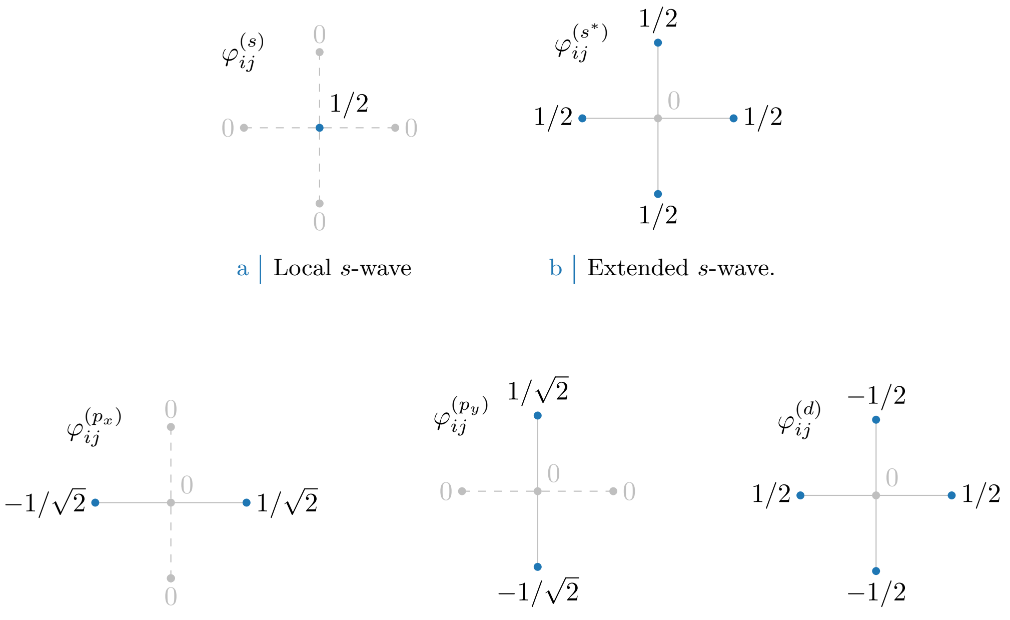
Local repulsion. A somewhat identical argument can be carried out for the local interaction,

$$\hat{H}_U = U\sum_{\mathbf{r}\in\mathcal{S}}\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\downarrow}$$

The only difference here is made by the fact that the interaction is repulsive on-site, thus the final structure factor in the Fourier transform disappears as well as the minus sign,

$$\hat{H}_U \overset{\mathcal{F}}{=} \frac{U}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{1.40}$$

When contracting these fermionic operators, identical considerations as for the non-local counterpart (1.34) hold. Note the similarity of this Fourier transform with the o.s. sector of the non-local attraction, given in Eq. (1.34). With this, we conclude the general introduction to the model and pass to the setup of the Hartree-Fock (HF) algorithm.



a | Local s-wave b | Extended s-wave.

c | p_x-wave. d | p_y-wave. e | $d_{x^2-y^2}$-wave.

Figure 1.2 | Form factors at different topologies, as listed in Tab. 1.5. In figures five sites are represented: the hub site i and its four NN. Solid lines represent non-zero values for $\varphi_{\boldsymbol{\delta}}$, while dashed lines represent vanishing factors.

1.4 Square lattice spatial symmetry channels

In next sections we introduce spatial symmetry structures and discuss their relation with point groups. We will aways refer to various spatial structures as "symmetry channels", throughout the system is able to establish a given order parameter defined in space.

Looking at $\hat{H}$, the main difference between its two-body interactions $\hat{H}_U$ and $\hat{H}_V$ of course relies in locality (first) and non-locality (second). Both operators in principle are able to break C_4 symmetry, as will later be discussed extensively.

For the 2D square lattice, various spatial symmetries are possible. Let i be a specific site, and consider a generic complex function g_{ij} whose domain is given by the site itself and its four NNs (the domain concides with the points of Fig. 1.1). The function can be written in the C_4 basis

$$g_{ij} = \sum_{\gamma} g^{(\gamma)} \varphi_{ij}^{(\gamma)}$$

where $g^{(\gamma)} \in \mathbb{C}$ are the g_{ij} symmetry channels coefficients, while $\varphi_{ij}^{(\gamma)}$ are the form factors listed in Tab. 1.5 and plotted in Fig. 1.3, a simple orthonormal rearrangement of the harmonics basis.

When dealing with phases which respect translational symmetry, or at most reduce it to a weaker symmetry by shrinking the Brillouin Zone, it's useful to work in reciprocal space, as discussed in Sec. **??**. As said, a particular phase can show specific spatial symmetries, and so will the Hartree-Fock parameters. The key point to notice is that self-consistency equations involve sums (or integrals) to be computed over fermionic states, which can be heavily simplified by employing said symmetries. Consider the form factors of Tab. 1.5, and their Fourier transforms of Tab. 1.6. They satisfy the relations:

$$\frac{1}{L_x L_y} \sum_{ij} \varphi_{ij}^{(\gamma)} \varphi_{ij}^{(\gamma)*} = \delta_{\gamma\gamma'} \qquad \frac{1}{L_x L_y} \sum_{\mathbf{k}} \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} = \delta_{\gamma\gamma'}$$

If these symmetries are developed by the HFPs, it's of no use to integrate over the entire BZ, since different regions linked by periodicity of the form factors will lead to redundant calculations.

Structure	Form factor	Graph
s-wave	$\varphi^{(s)}_{\mathbf{rr'}} = \delta_{\mathbf{rr'}}$	Fig. 1.2a
Extended s-wave	$\varphi^{(s^*)}_{\mathbf{rr'}} = \frac{1}{2}\left[\delta_{\mathbf{r'}=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r'}=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r'}=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r'}=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 1.2b
p_x-wave	$\varphi^{(p_x)}_{\mathbf{rr'}} = \frac{1}{\sqrt{2}}\left[\delta_{\mathbf{r'}=\mathbf{r}+\underline{\mathbf{x}}} - \delta_{\mathbf{r'}=\mathbf{r}-\underline{\mathbf{x}}}\right]$	Fig. 1.2c
p_y-wave	$\varphi^{(p_y)}_{\mathbf{rr'}} = \frac{1}{\sqrt{2}}\left[\delta_{\mathbf{r'}=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r'}=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 1.2d
$d_{x^2-y^2}$-wave	$\varphi^{(d)}_{\mathbf{rr'}} = \frac{1}{2}\left[\delta_{\mathbf{r'}=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r'}=\mathbf{r}-\underline{\mathbf{x}}} - \delta_{\mathbf{r'}=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r'}=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 1.2e

Table 1.5 | Form factors for a function defined on a square lattice, with a prefactor included for normalization. We use lattice vector notation here. The last column indicates the graph representation of each contribution given in Fig. 1.2. Subscript $x^2 - y^2$ is omitted for notational clarity.

Structure	Structure factor	Graph
s-wave	$\varphi^{(s)}_{\mathbf{k}} = 1$	Fig. 1.2a
Extended s-wave	$\varphi^{(s^*)}_{\mathbf{k}} = \cos k_x + \cos k_y$	Fig. 1.2b
p_x-wave	$\varphi^{(p_x)}_{\mathbf{k}} = i\sqrt{2}\sin k_x$	Fig. 1.2c
p_y-wave	$\varphi^{(p_y)}_{\mathbf{k}} = i\sqrt{2}\sin k_y$	Fig. 1.2d
$d_{x^2-y^2}$-wave	$\varphi^{(d)}_{\mathbf{k}} = \cos k_x - \cos k_y$	Fig. 1.2e

Table 1.6 | Structure factors derived from the correlation structures of Tab. 1.5. The functions hereby defined are orthonormal, and are plotted in Fig- 1.3.

Finally, a very useful detail is given by the following argument. Consider the equation

$$\cos(\delta k_x) + \cos(\delta k_y) = \cos k_x \cos k'_x + \sin k_x \sin k'_x + \cos k_y \cos k'_y + \sin k_y \sin k'_y$$

For the sake of readability, the notations

$$c_\ell \equiv \cos k_\ell \qquad s_\ell \equiv \sin k_\ell \qquad c'_\ell \equiv \cos k'_\ell \qquad s'_\ell \equiv \sin k'_\ell$$

are used. Group the four terms above,

$$\underbrace{\left(c_x c'_x + c_y c'_y\right)}_{\text{Symmetric}} + \underbrace{\left(s_x s'_x + s_y s'_y\right)}_{\text{Anti-symmetric}} \tag{1.41}$$

The first two exhibit inversion symmetry for both arguments $\mathbf{k}$, $\mathbf{k}'$; the second two exhibit anti-symmetry. Decoupling the symmetric part, we obtain

$$c_x c'_x + c_y c'_y = \frac{1}{2}(c_x + c_y)(c'_x + c'_y) + \frac{1}{2}(c_x - c_y)(c'_x - c'_y)$$

which finally gives:

$$\begin{aligned}
\cos(\delta k_x) + \cos(\delta k_y) = & \frac{1}{2}(c_x + c_y)(c'_x + c'_y) && (s^*\text{-wave}) \\
& + s_x s'_x && (p_x\text{-wave}) \\
& + s_y s'_y && (p_y\text{-wave}) \\
& + \frac{1}{2}(c_x - c_y)(c'_x - c'_y) && (d_{x^2-y^2}\text{-wave})
\end{aligned}$$

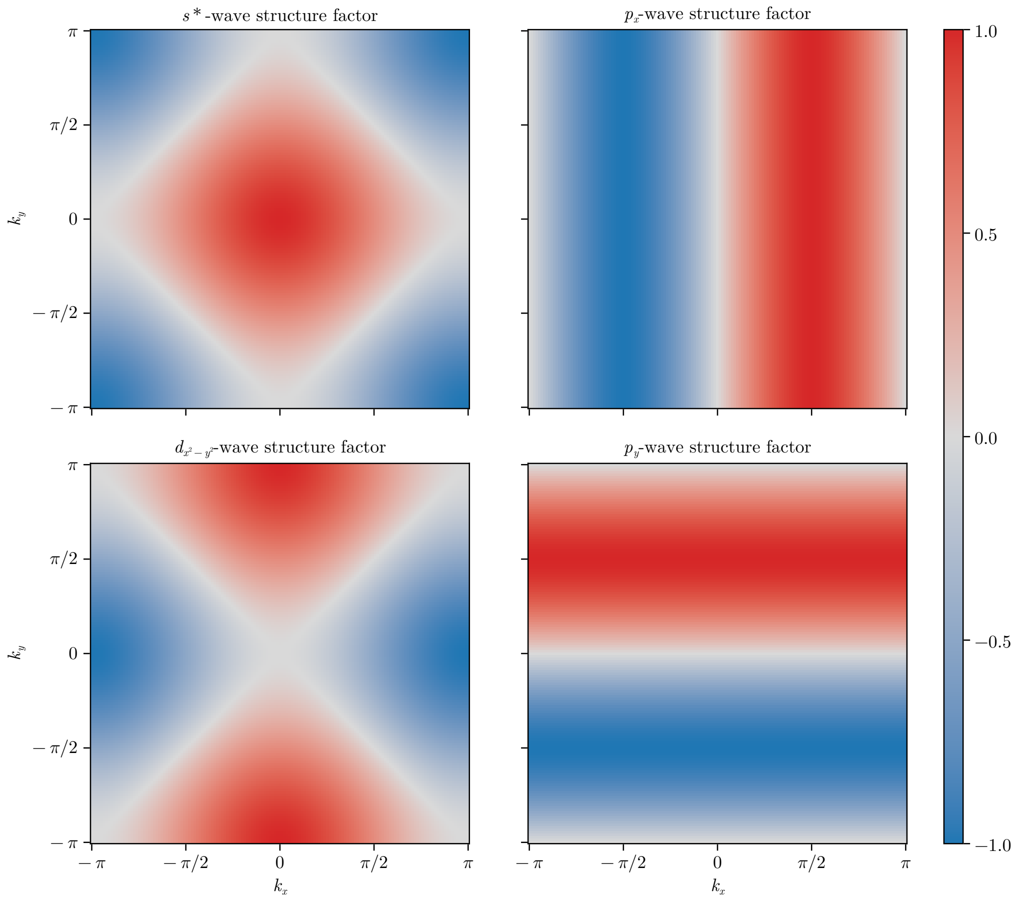


Figure 1.3 Representation of the various structure factors listed in Tab. 1.6 on the BZ. For the s^* and $d_{x^2-y^2}$-wave factors, the real part is plotted; for the p_ℓ factors, the imaginary part is plotted.

or, in a more compact form

$$\cos(\delta k_x) + \cos(\delta k_y) = \frac{1}{2}\sum_\gamma \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}'}^{(\gamma)*} \quad \text{for } \gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\} \tag{1.42}$$

$$= \frac{1}{2}\sum_\gamma \varphi_{\mathbf{k}}^{(\gamma)*} \varphi_{\mathbf{k}'}^{(\gamma)} \tag{1.43}$$

This decomposition will become particularly handy in later calculations.

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